

Credit Risk Analysis Using SQL

Understanding loan default patterns
to support smarter lending decisions

32,581

Loan Records

12

Raw Features

~22%

Default Rate

What This Report Covers

01

Business Problem

Why loan default matters and project goal

02

Key Questions

Five analytical questions driving the analysis

03

Dataset Overview

32,581 applications, 12 variables + engineered metrics

04

Data Quality

Missing values, outliers, cleaning audit

05

Methodology

Six-step analytical pipeline

06

Findings & Insights

Default rates by grade, income, intent, age, DTI

07

Recommendations

Business Problem

The Problem

Lenders face significant financial losses when borrowers default. To manage risk effectively, they need a clear understanding of which applicant characteristics signal higher likelihood of default.

Project Goal

Use SQL-based analysis to uncover the strongest predictors of loan default and translate them into actionable insights for improved lending decisions.

32,581

Loan applications analysed

~22%

Overall default rate in dataset

6

Key risk dimensions investigated

5

Actionable recommendations produced

Key Analytical Questions

01**How reliable is the dataset?**

Missing value audit + outlier checks before analysis

02**Which borrower characteristics influence default?**

Age, income, home ownership, employment length

03**How strongly does loan grade predict default?**

Grade A → 90% paid; Grade G → 98% default

04**Which loan purposes carry higher risk?**

Venture, Education and Personal loans top the list

05**How do DTI & LTI relate to default?**

High-DTI borrowers default at 3× the rate of Low-DTI

Dataset Overview

32,581

Total Records

12

Raw Features

3

Engineered Metrics
(LTI, DTI, Age Group)

~22%

Default Rate

Borrower Demographics

Age · Income · Home Ownership · Employment Length

Loan Characteristics

Amount · Grade · Purpose · Interest Rate

Credit Behaviour

Credit History Length · Prior Default on File

Engineered Features

Loan-to-Income (LTI) · DTI Risk Category · Age Group

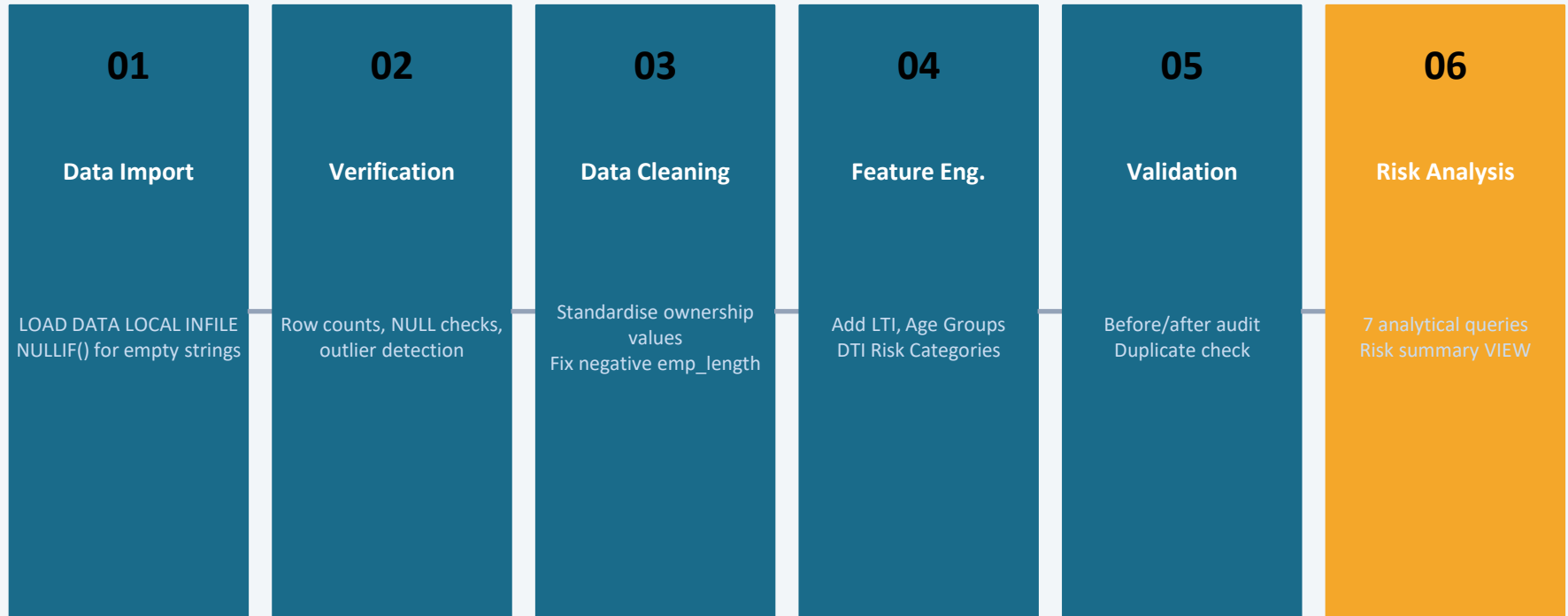
Target Variable

Loan Status: 0 = Default · 1 = Paid

Data Quality Check Summary

Check Performed	Result	
Total Records Loaded	32,581 records loaded into credit_risk table	✓
Missing Values	895 missing employment lengths · 3,116 missing interest rates — converted to NULL on import	✓
Invalid Age / Income / Rate	0 records with age ≤ 0 or > 120 · 0 with negative income · 0 with rate outside 0–100%	✓
Empty Strings → NULL	All empty strings successfully converted to NULL via NULLIF() during LOAD DATA	✓
Duplicate Records	Small number detected but volume was insignificant — no action required	✓
Zero-Income After Cleaning	0 records with zero income in credit_risk_clean	✓
Dataset Size Before vs After	Before: 32,581 · After: 32,581 (no rows removed — cleaning was in-place via CASE logic)	✓

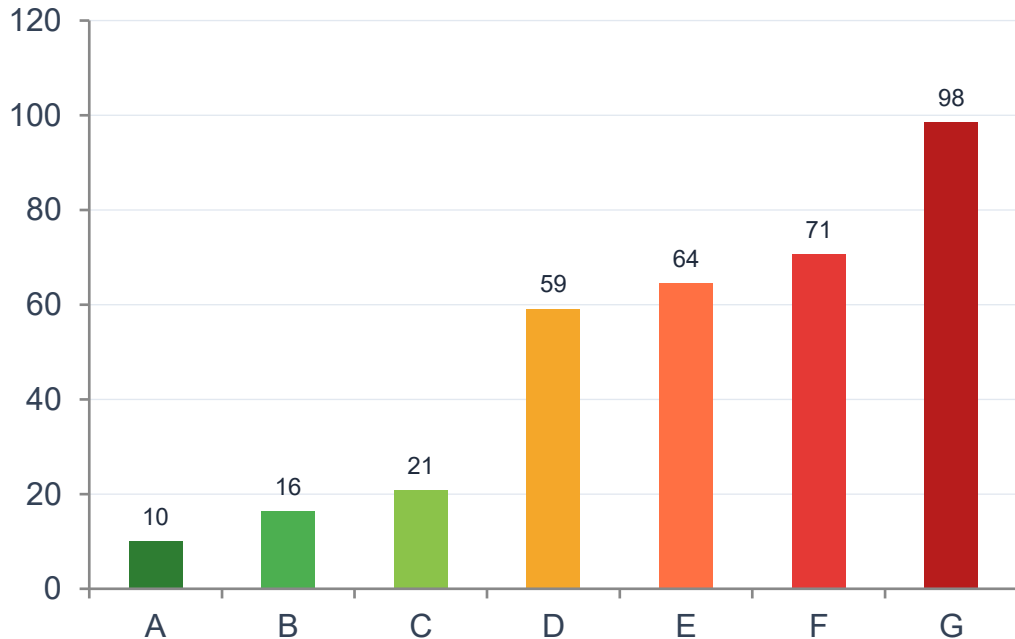
Methodology — Six-Step Pipeline



Each step produces a verifiable output — ensuring the analysis is reproducible and auditable from raw CSV to final risk view.

Risk by Loan Grade — Strongest Predictor

Default Rate (%) by Loan Grade



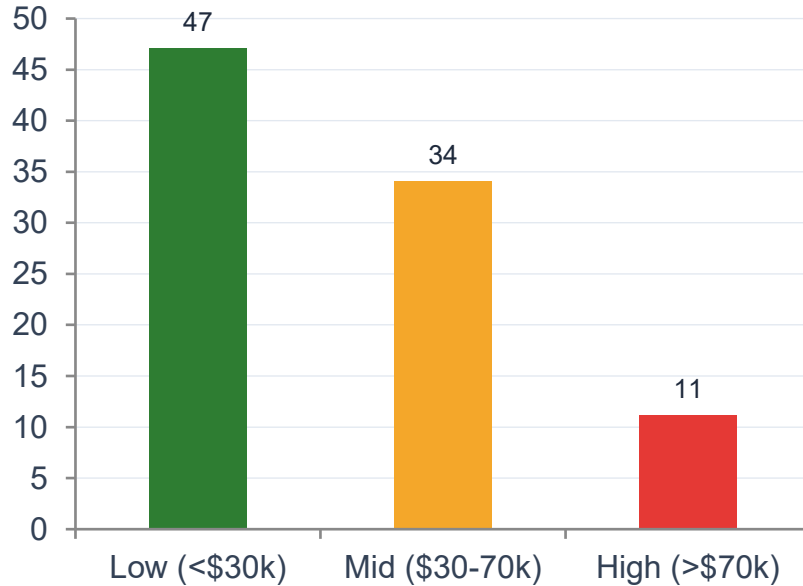
Grade	Volume	Default%	Avg Rate
A	10,777	9.96%	7.33%
B	10,451	16.28%	11%
C	6,458	20.73%	13.46%
D	3,626	59.05%	15.36%
E	964	64.42%	17.01%
F	241	70.54%	18.61%
G	64	98.44%	20.25%



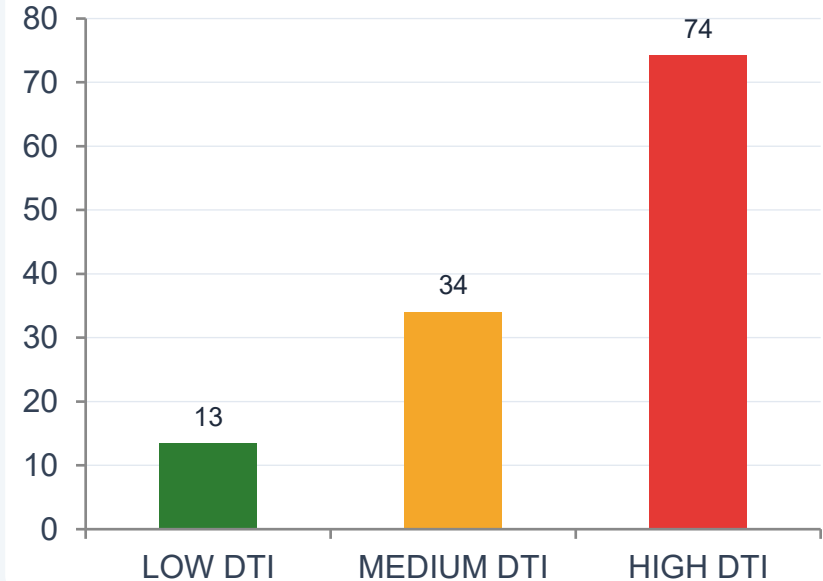
Loan grade is the single strongest predictor — Grade G borrowers default at 98.4% vs just 9.96% for Grade A.

Income Level & DTI Risk — Financial Capacity Drives Default

Default Rate (%) by Income Bracket



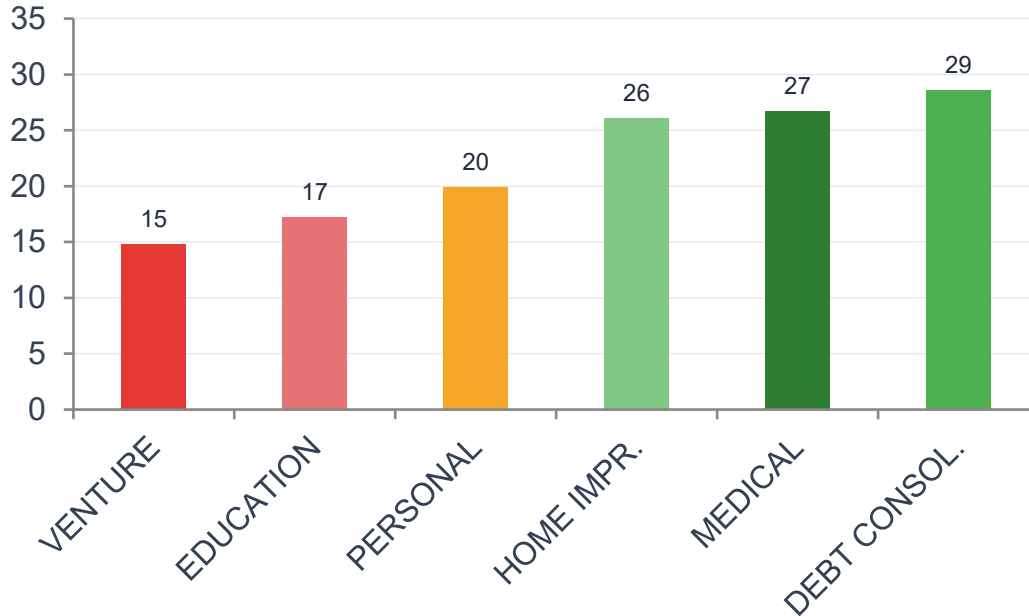
Default Rate (%) by DTI Risk Category



💡 Low-income borrowers default at 47% vs 11% for high-income. High-DTI applicants default at 74% — nearly 3× the Low-DTI rate of 13%.

Loan Intent & Home Ownership Risk

Default Rate (%) by Loan Intent



Home Ownership × Employment Tenure

Ownership	Tenure	Default%
OWN	Senior (2yr+)	6.32%
OWN	Junior (0-2yr)	11.81%
MORTGAGE	Senior (2yr+)	12.18%
MORTGAGE	Junior (0-2yr)	14.96%
OTHER	Senior (2yr+)	25.00%
RENT	Senior (2yr+)	30.26%
RENT	Junior (0-2yr)	34.99%
OTHER	Junior (0-2yr)	41.03%

💡 Venture and Education loans carry the highest default risk (14–29%). Renters with short job tenure are the riskiest ownership segment at 35–41%.

Key Risk Insights

01

Loan Grade — Strongest Predictor

Grade G: 98.4% default vs Grade A: 9.96%. A 10× difference — loan grade must anchor every risk model.

02

Financial Capacity Drives Default

Low-income borrowers default at 47% vs 11% for high-income. High-DTI applicants default at 74% vs 13% for Low-DTI.

03

Stability Indicators Matter

Renters with junior employment (0-2yr) default at 35-41%. Homeowners with senior tenure default at just 6-12%.

04

Loan Purpose Signals Intent Risk

Venture (14.8%) and Education (17.2%) loans show highest default. Debt Consolidation is the lowest risk intent at 28.6%.

05

Younger Borrowers Default More

18-24 age group carries the highest default rate. Risk decreases steadily with age as financial maturity develops.

Recommendations — Risk-Driven Lending Actions

1

Prioritise Loan Grade

Make loan grade central in approval scoring. Grade D and below carry 59%+ default rates — require stronger compensating factors.

2

Tighten DTI & Income Checks

High-DTI borrowers (>40% income to debt) default at 74%. Apply stricter income verification and DTI caps.

3

Factor in Stability Indicators

Renters with <2yr employment are 3× riskier than homeowners with senior tenure. Weight home ownership + tenure in scoring.

4

Scrutinise High-Risk Loan Intents

Apply stricter review to Venture and Education loans. Consider lower approval limits or higher rate adjustments for these intents.

5

Adjust Thresholds by Age Group

Younger borrowers (18-24) show the highest default rates. Require stronger collateral or co-signers for applicants in this age band.

Thank You

for your Attention

Gerald Nwachukwu

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At a Glance

98.4%

Grade G default rate

74%

High-DTI default rate

47%

Low-income default rate

35%

Renters (junior) default rate

6

Risk factors identified